

Second Bite

A stylized green carrot icon with a leafy top, positioned between the words 'Bite' and 'e' in the title 'Second Bite'.

London's AI-powered food redistribution solution

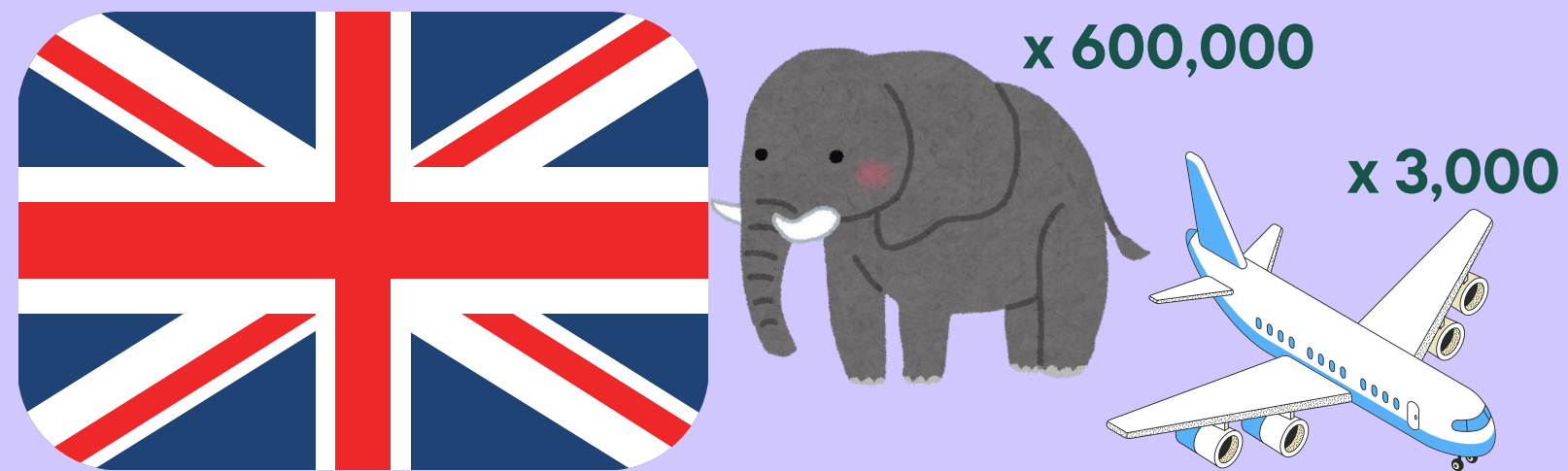
MG4J8 - Week 11 Final Report

Amaan A, Jasmine M, Alicia R, Naoise L, Robbie K, Coen TW, Zane Q

Part I



Context and Solution



2.9 million tonnes of commercial food is wasted every year

01 Majority of waste from restaurants and supermarkets

02 Food insecurity is a major issue
Food donation process lacks clarity, efficiency, and incentives

03 Technology gap to optimize the space for all parties involved

Our Solution



01

Multi-sided AI-Platform connecting charities and donors

02

Diverse algorithmic architecture optimizing every step of donation process

03

Decrease food waste from restaurants and supermarkets

04

Increase donation consistency for charities and food banks

05

Incentivize donor participation through streamlined tax incentives

~10 million tonnes of CO²
Feed **15 million** Londers

Business Model



Value Creation

Value for Donors

- Optimised tax savings
- Streamlined donation (decision-making) process
- Meet SDG commitments

Value for Charities

- Greater donation volumes
- More certain supplies

Monetization

- 5%-10% charge of donors' tax savings
- Charities and donor membership fee
- Premium insights services (e.g. donor inventory mgmt insights)

Launch & Growth

- Launch in London for critical mass
- Manage network effects by subsidizing donors

Competitive strategy:

- Focus on tax savings
- Build to become industry-wide coordination platform

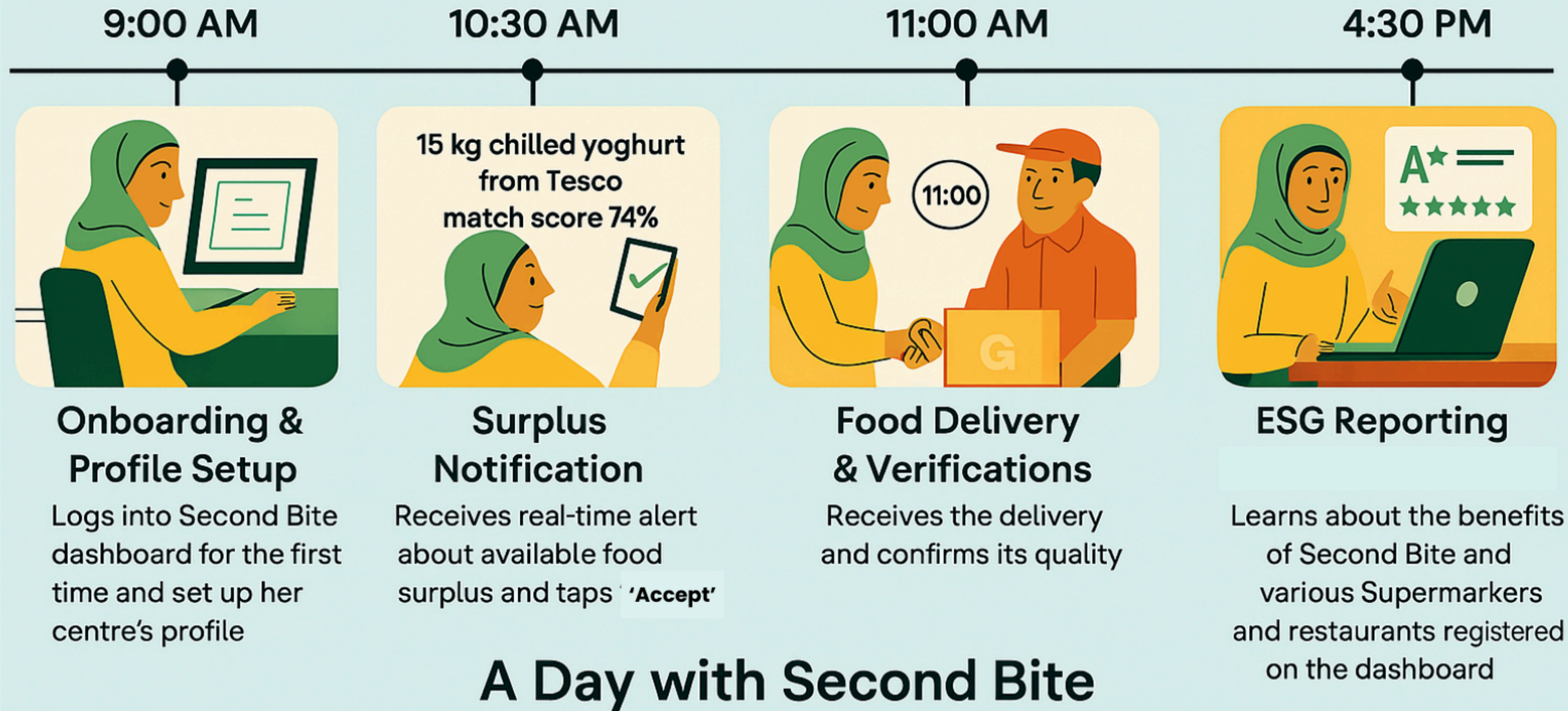
Part II



Technological Solution

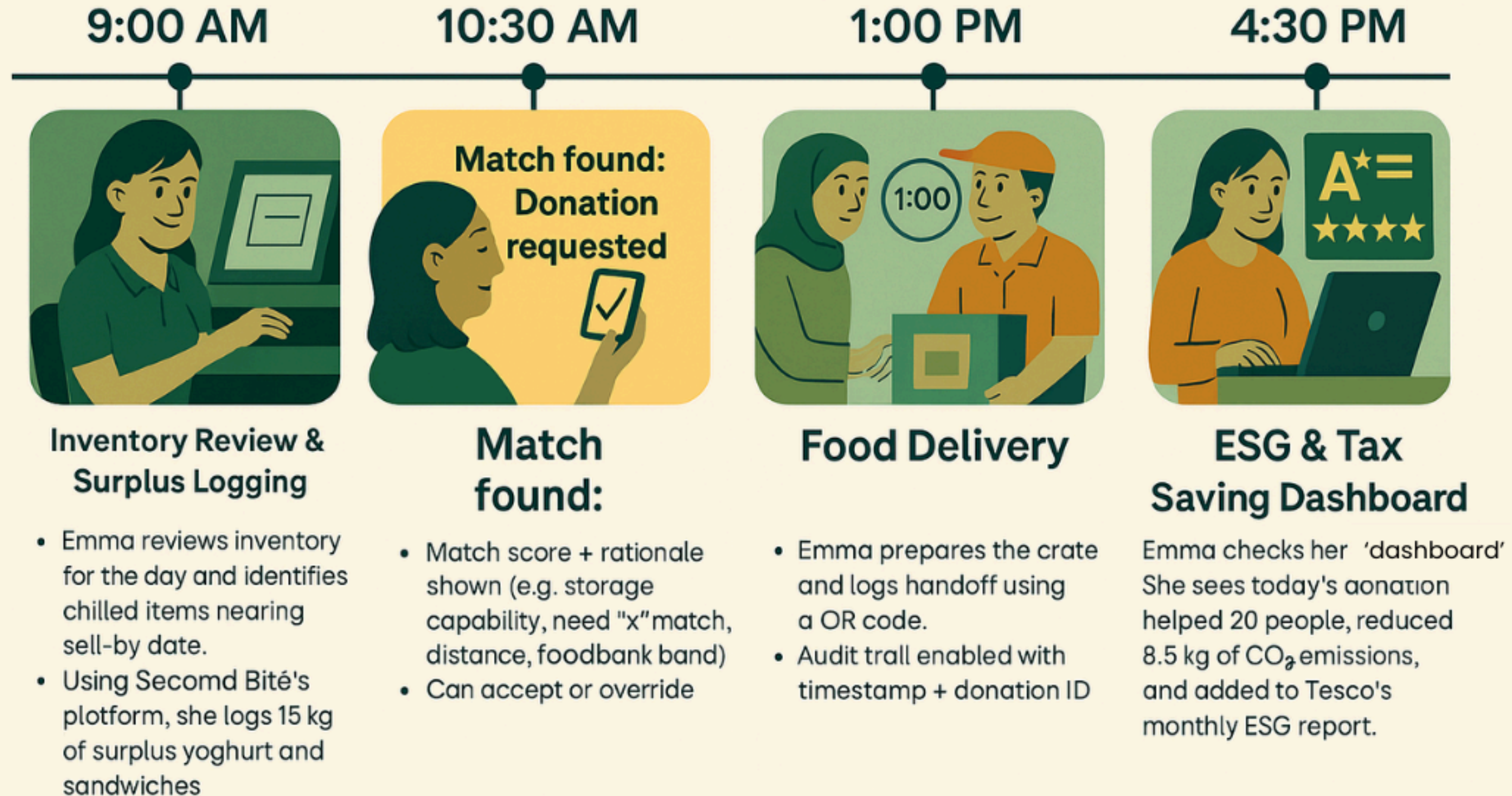
Charity User Journey:

A Day in the Life of Sarah Malik



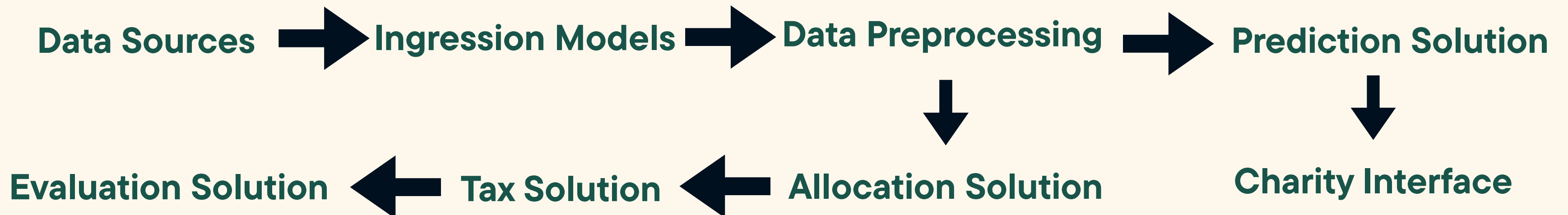
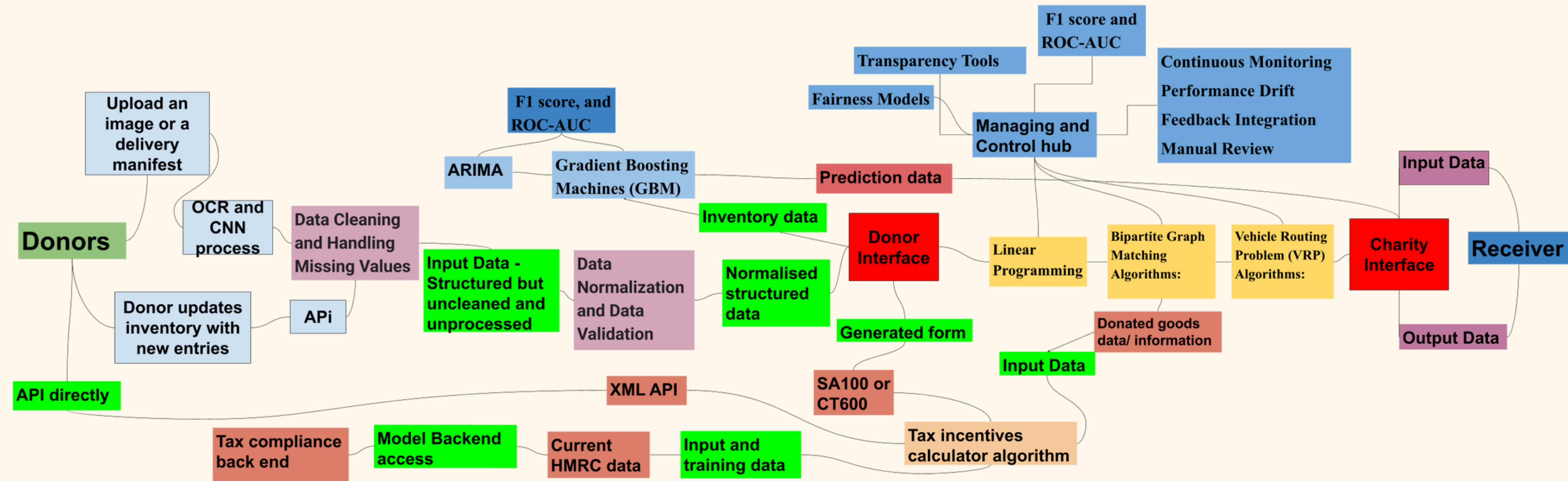
Donor User Journey:

A Day in the Life of Emma-Jane

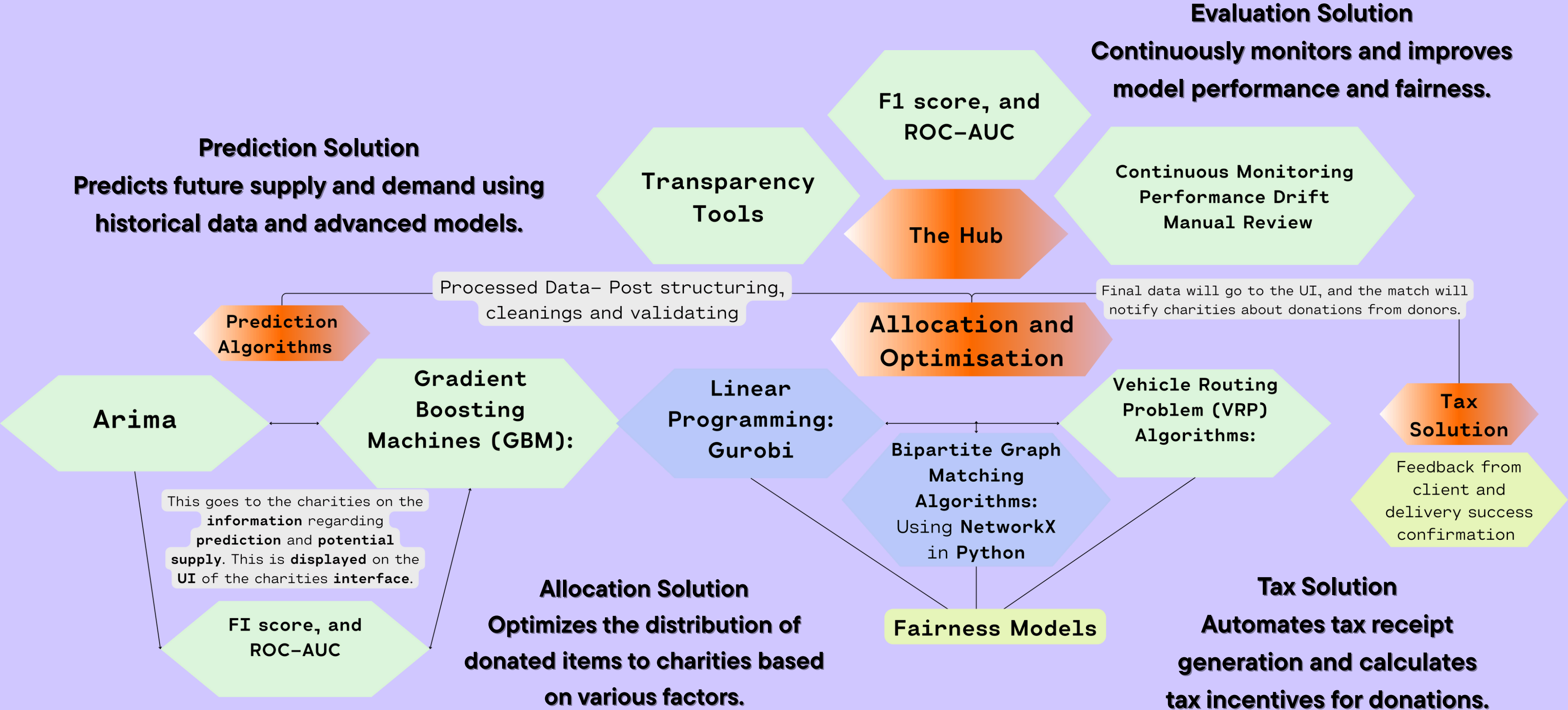


A Day with Second Bite

Technology Solution Overview



Allocation and Matching Solution Overview



Part III — Organisational Solution

Organisation



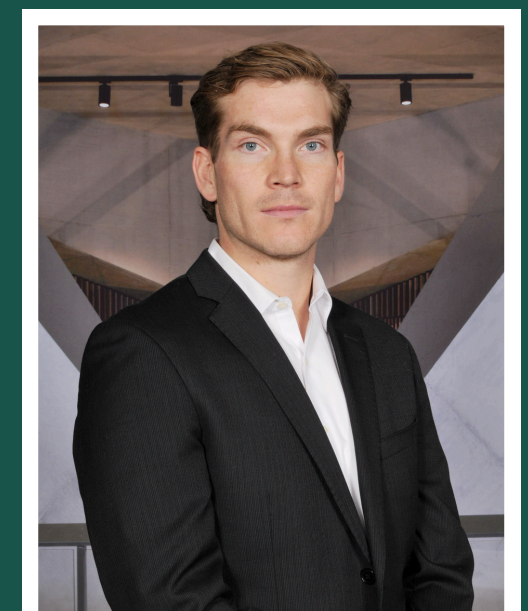
Naoise - CTO



Coen - CEO



Alicia - CFO



Robbie - COO



**Amaan - Lead
Technical Designer**



**Zane - Donor
Manager**



**Jasmine - Charity
Manager**

Identity



Social Enterprise, not a Tech Company

- Core focus on solving social problem using technological innovation

Tech in Service of Social-Enterprise Goals

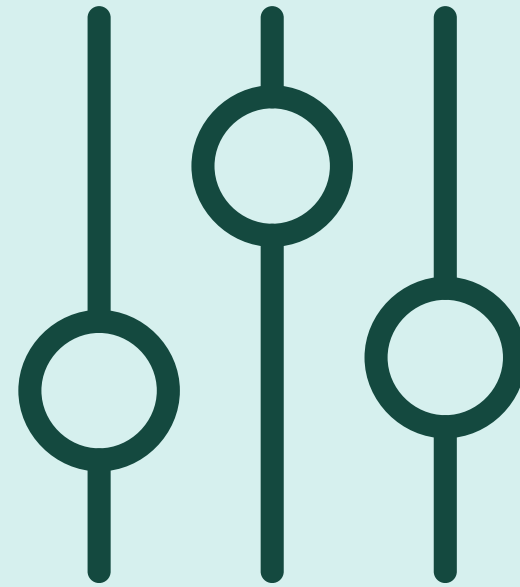
- Social goals encoded in AI systems
- Donor and Charities Managers lead model improvement work

Human-AI Engagement

Trust



Control

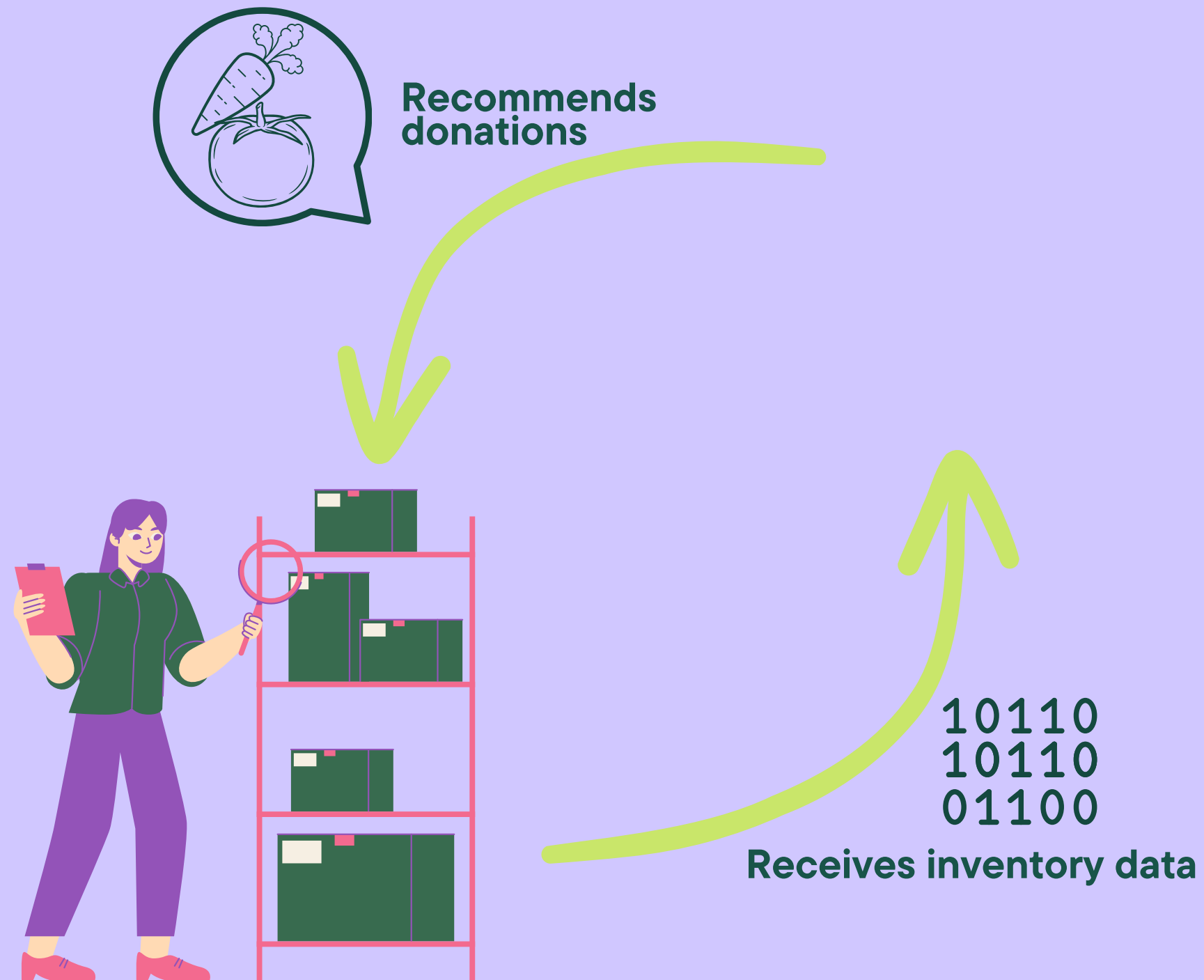


Transparency



Organisational Impact for Stakeholders

Potential Work Systems & Role Boundary impact



Training for Effective Human-AI Engagement



Monetisation Strategy

Subscription Fees

We will charge a nominal fee to charities and Donors to help pay for the upkeep of the site

Tax Write Off Percentage

We will Charge between 5%-10% of the tax savings that we provide to our donor companies

- Self Assessment Tax Return (SA100)
- Corporation Tax Return (CT600)

Leverage Collected Data

Offering access to our prediction data, Wastage data and charity network ethically and responsibly

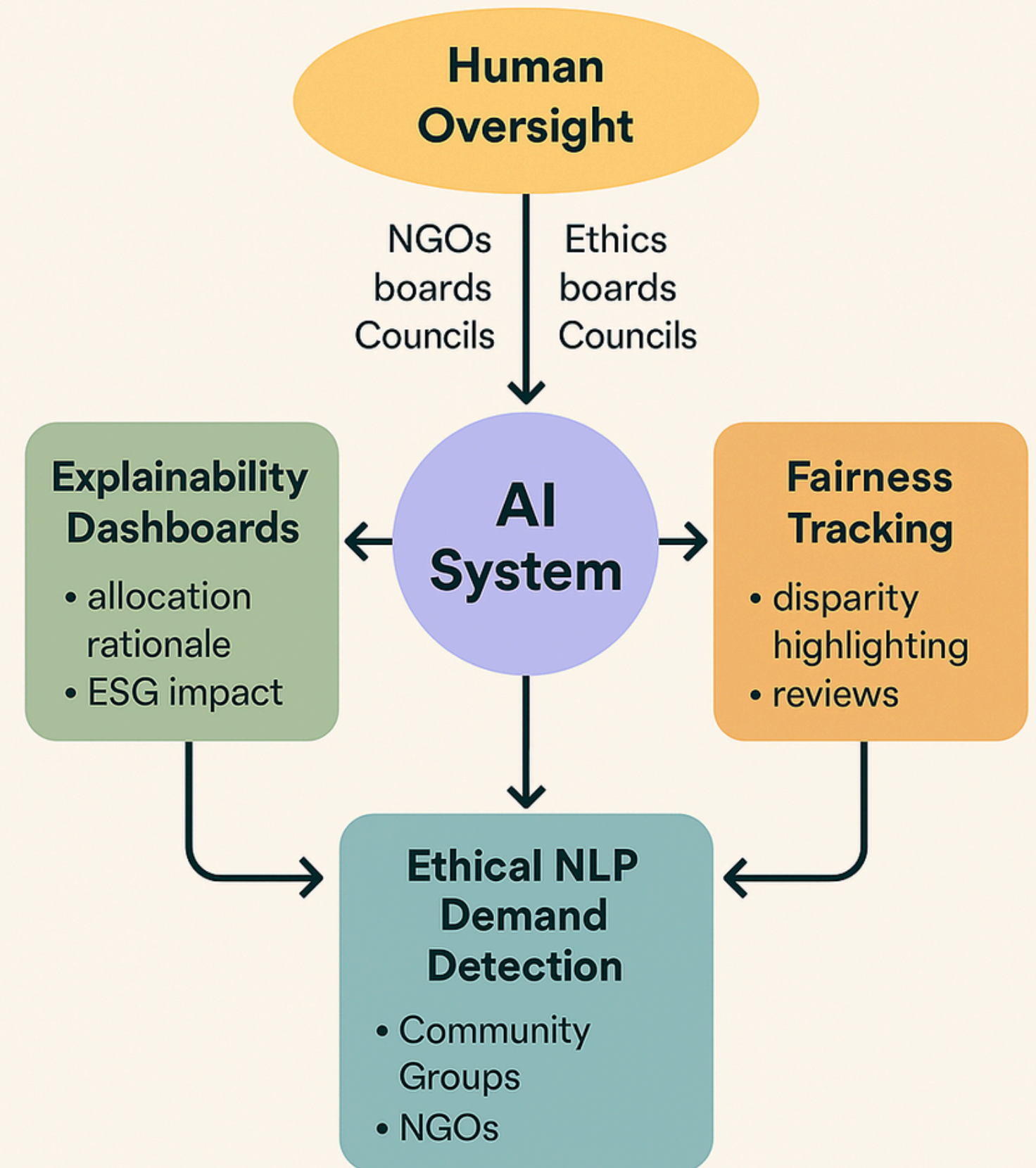
Part IV — Governance &
Legal
Considerations

4 Pillars of AI Governance Design



- Our governance map ensures that Second Bite's AI system is guided by transparent, accountable, and equitable decision-making
- Each actor plays a defined role in monitoring, contesting, and improving the AI's behaviour in real time.

Stakeholder Governance Map



Important Regulations & Implications



ISO 27001: Key for EU expansion

Data Privacy and Security

GDPR
NIST AI RMF

NLP models must use only aggregated, excluding personal identifiers. Partner NGO records must be deleted after 6 months without explicit consent.

Food Safety Rules: Temp logs ($\pm 2^{\circ}\text{C}$), handover docs, no spoilage.

Food Standards

UK FSA (1990)
UK Food Standard Agency (1999)
DEFRA

Tax Benefits

HMRC Tax Compliance
Charities 2016
ISO 27001

AI calculates tax deductions based on:
Expiry, Nutritional value and Market alternatives

Bias



Bias in **data** and **design**

- Reinforced patterns
- Convenience over need
- Data collection gaps and underrepresentation



Accountability

Who is responsible for missed deliveries, food waste, or exclusion?

- donors, charities, drivers, software developers, algorithms
- The

In our system, decisions are made across multiple layers:

- Prediction, allocation, & route optimisation
- Data layers introduce new decision points
- traceability to identify failure

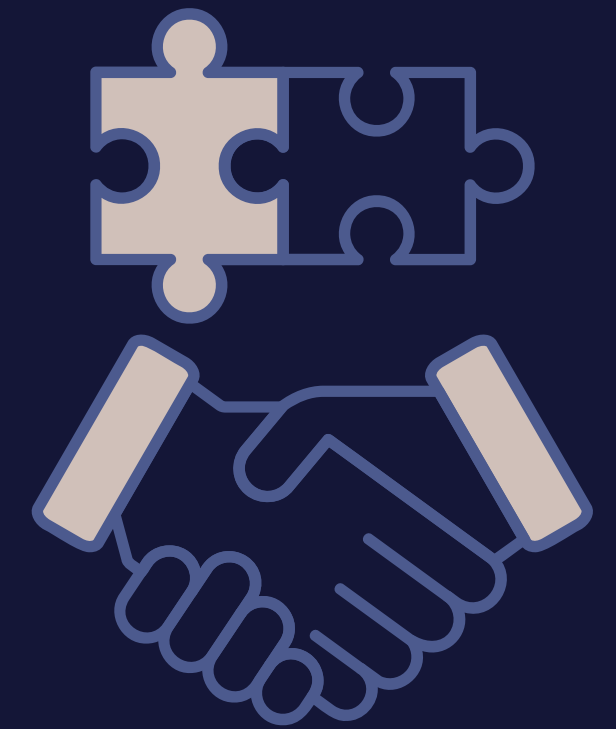
Responsibility diffusion:

- No one actor can take ownership over outcomes that harm/exclude communities



Transparency

- In AI-driven systems, decisions can be opaque
 - Effects food donation when charities:
 - Don't know why they haven't received food when they usually do?
 - Don't know why certain food types get allocated without alignment with dietary or cultural needs?
 - **Power asymmetry**.
 - Those who design or manage the system have visibility and control
 - Those on the receiving end—often with fewer resources—are expected to accept outputs without recourse.



Part V — **Social & Ethical
Considerations**

Ethical Challenges



Bias

- **Delivery convenience** > **actual need**
- **Reinforced patterns** - exclusion of small actors
- **Data collection** - supermarket APIs vs manual entries

Accountability

- Missed deliveries, food waste, exclusion - **who is responsible?**
- Multiple **data layers** & **decision points**
- **Diffused responsibility** - accountability gap

Transparency

- **System opacity**
- Only those who **design/manage** the system **have control**
- **Power asymmetry** - users are forced to accept decisions

Cybersecurity

- **Logistics APIs** could expose sensitive data
- **Internal misuse**
- **Metadata leaks** - profiling of vulnerable communities

4 Pillars of AI Governance Design



Human Oversight

NGOs approve, reject, or reallocate AI decisions

All responses fed into reinforcement learning model

Human-in-the-loop + improving model responsiveness

Explainability Dashboards

Visualized rationale:

- Allocation score
- Transport constraints
- Expiry timeline & storage match

ESG impact summaries:

- # meals delivered
- GHG emission reduction
- Waste reduction (tonnes saved)

Fairness Tracking

Dashboards highlight disparities

Monitors over/under-allocation of NGOs or regions

Triggers governance reviews & prompts human intervention

Ethical NLP Design Demand Detection

Social media scraping done via public, anonymized datasets

BERT-based model tuned for community reporting trends

Validation loop with councils and on-the-ground NGO feedback to confirm inferred demand

Ethical Framework



Bias

- **Fairness metrics** to assess allocation outcomes (across groups)
- **Need-based allocation** - incorporate food insecurity into allocation logic
- **Inclusive data collection** - entry tools (for small charities)

Accountability

- **Explainable allocation** - provide clear justifications (of matches)
- **Human in the loop** - flag high-risk decisions (for review)
- **Feedback mechanism** - collect feedback from users (after each allocation)

[Donor Input]



[Data Ingestion] → [Inclusive data collection ✓]



[Allocation Engine]



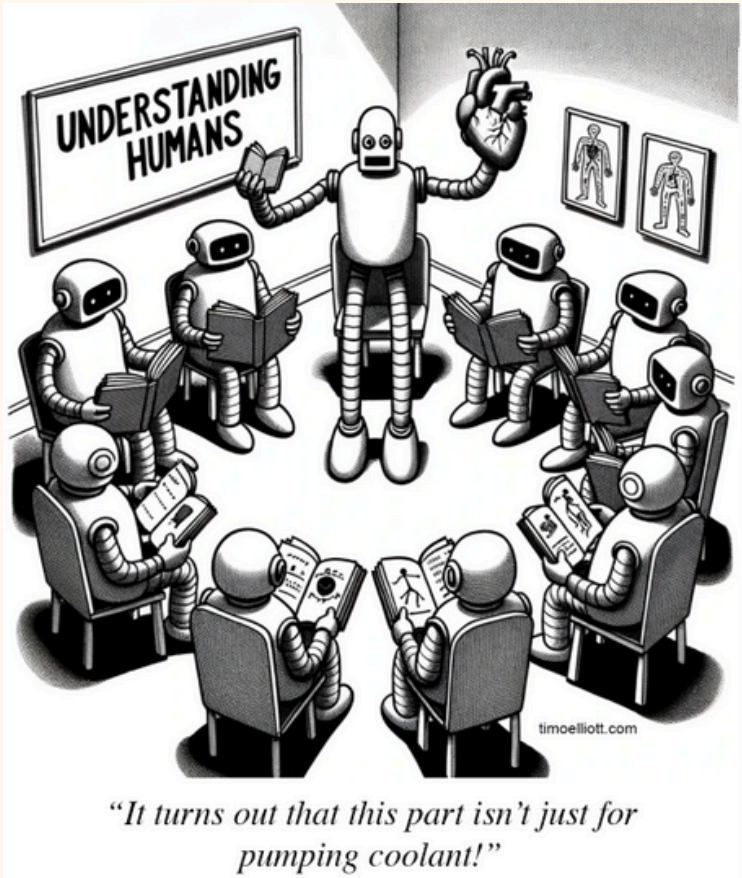
→ [Fairness Metrics ✓]

→ [Explainable Match Justification ✓]

→ [Human Review if high-risk ✓]



[Feedback Loop] → [Transparency + Accountability ✓]



Ethical Framework



Transparency

- **Allocation justification** - explain why donations were/weren't allocated (to a charity)
- **Decision logs** - key decision steps/timestamps (for each allocation)
- **Feedback dashboard** - charities can track outcomes of their feedback

Cybersecurity

- Use **secure, tokenised APIs** to encrypt data
- Apply **role-based access controls** (RBAC)
- Implement **data minimisation**



The BiteRight Shield

Uses tokenised APIs, role-based access control, and data minimisation to safeguard sensitive information and build trust.

Training & Adoption



Food Aid Wholesale Distributors (FAWDs)

Training:

- Required for new 2ndByte interfaces.
- Online training for staff involved in collection, processing, and delivery.

Adoption:

- Integrate new system into existing workflows.
- Supervise model predictions to ensure accuracy and efficiency.

Food Aid Service Organisations (FASOs)

Training:

- Focus on operational and organizational adoption formally
- Online training to adapt to automated processes.

Adoption:

- Implement automated processes
- Utilize direct connectivity to inventory management

Donors

Training:

- Decision-making based on financial-benefit overviews.
- Online training for new interfaces and financial-benefit overviews.

Adoption:

- Automate donor indications.
- Use financial-benefit overviews to make informed decisions

Confucius

“He who wishes to
secure the good of
others, has already
secured his own.”





Q/A

Second Bite 

THANK YOU



Second Bite 